REVISIONS

SYM	DESCRIPTION	DATE	APPROVAL

ITEM	QTY	PART NO.	DESCRIPTION	SPECIFICATION
1	2	Stk. No. 626-0040	Fitting, Comp - Male Conn.	Brass 1/8NPT x 3/8 tube
2	4	626-0041	Fitting, Comp - Male Conn.	Brass 1/4NPT x 3/8 tube
3	4	1413-0006	Nozzle, Hose	Brass 3/8 hose x 1/4NPT
4	1	575-0011	Hose, Rubber	3/8 I.D. 2 Ply x 16 Ft. Lg.
5	4	Stk. No. 331-0002	Clamp, Hose	Brass 3/4 I.D.
6	2		Tubing	Copper 3/8 O.D. x 18" Lg.
7	1		Photo Bulletin No. 4	

LIST OF MATERIALS

NEXT ASSY	USED ON
APPLICATION	

UNLESS OTHERWISE SPECIFIED
DIMENSIONS IN INCHES
TOLERANCE ON:

FRACTIONS	DECIMALS	ANGLES
±	±	±

DRAWN BY	DATE
CHECKED w. Fisher	DATE 8-13-74
APPROVED FOR MFG	DATE
ENGINEER	DATE

NAME

CHART LOAD COIL
TRANSFORMER PLUMBING
PARTS
(LCT-4 WITH GENERATOR)

MATERIAL

FINISH

LEPEL
HIGH FREQUENCY LABORATORIES, INC.
59-21 QUEENS-MIDTOWN EXPRESSWAY
MASPETH, NEW YORK CITY

DWG SIZE	DRAWING NO.	REV
A	3350013	

SCALE

SHEET OF

OPERATING HAZARDS

READ THE FOLLOWING INSTRUCTIONS AND TAKE ALL THE NECESSARY PRECAUTIONS.

OPERATING INSTRUCTIONS

This information is provided to help you establish safe operating conditions for both you and your ITT Electron Tube.

Use the Tube Data Sheet and Operating Instructions with the information given in this sheet to help you operate this tube in a safe and efficient manner. The Tube Data Sheet provides the general characteristics and maximum ratings for the tube. The Operating Instructions give special considerations and precautions to be followed to obtain best performance.

Do not operate this tube except in accordance with proper operating instructions, these precautions, and any additional information provided by ITT Electron Tube Division representatives. Address any questions regarding the safe and proper use of this tube to:

ITT Electron Tube Division
P.O. Box 100
Easton, Pennsylvania 18042

WARNING—SERIOUS HAZARDS EXIST IN THE OPERATION OF ALL TUBES

The operation of all electron tubes involve the following hazards:

a. HIGH VOLTAGE—Normal operating voltage can be deadly.
b. ELECTROMAGNETIC RADIATION—Electromagnetic radiation can cause serious personal injury which can be fatal.
c. X-RAY RADIATION—High Voltage tubes can produce dangerous X-rays.

Read the following instructions and take all necessary precautions. ITT, as a component supplier, can assume no responsibility for any damage or injury resulting from the operation of ITT Electron Tubes.

HIGH VOLTAGE

Operating voltages for electron tubes range from about 200 volts to over 150 kilovolts. Since these voltages can be deadly, the equipment must be designed properly and operating precautions must be followed. Design equipment so the operator cannot come in contact with high voltages. Enclose high voltage circuits and terminals and provide interlocking switch circuits to open the primary circuits of the power supply and discharge high voltage condensers whenever access is required.

ELECTROMAGNETIC RADIATION

Exposure of the human body to excess electromagnetic radiation is unsafe. For this reason, the rf energy must be contained properly by waveguides and shielding. Arrangements should be made to prevent exposure of personnel to strong rf fields in the vicinity of electron tubes and in front of antenna systems. (Ref.: Proc. IRE, Vol. 49, No. 2, pp. 427-447, Feb. 1961)

X-RAY RADIATION

Electronic tubes operating at voltages higher than 10 kilovolts produce progressively more dangerous X-ray radiation as the voltage is increased. Therefore, many high power electron tubes are potential X-ray hazards. Provide adequate X-ray shielding on all sides of these tubes, as well as the modular and pulse transformer tanks. Make periodic checks on the X-ray levels and never operate high voltage tubes without adequate X-ray shielding being in place. (Ref.: "Medical X-ray Protection up to Three Million Volts." National Bureau of Standards Handbook 76. Available from Superintendent of Documents, Washington, D. C. 20402.)

DANGER—BERYLLIUM OXIDE CERAMICS AVOID BREATHING DUST OR FUMES

Some ITT high power electron tubes contain Beryllium Oxide Ceramics and as such are clearly identified. Avoid performing any operation on the ceramic parts of these tubes which produce dust or fumes. In particular, avoid operations such as grinding, grit blasting, and acid cleaning. BERYLLIUM OXIDE DUST AND FUMES ARE HIGHLY TOXIC AND BREATHING THEM CAN RESULT IN SERIOUS PERSONAL INJURY OR DEATH.

Disposal of used tubes containing Beryllium Oxide must be done in a manner which will ensure that personnel in the disposal operation will be safeguarded and that personnel in any possible salvage operation will be fully warned. If proper disposal presents a problem, ITT is prepared to offer disposal service for tubes containing this material. Tubes should be returned prepaid with a written authorization to ITT to dispose of the tubes.

ELECTRON TUBE DIVISION ITT

ELECTRON TUBE DIVISION
P.O. Box 100
Easton, Pennsylvania 18042
Telephone 215 252-7331

Power Tube Applications

READ THOROUGHLY

Maximum Ratings—Maximum ratings are limiting values above which the serviceability of the tube may be impaired with respect to life and to satisfactory performance. The equipment designer has the responsibility of determining an average design value sufficiently below each maximum rating that the maximum values will never be exceeded under any condition of supply voltage variation or of load variation.

ITT power tube applications engineers are available to discuss any question which may arise concerning maximum ratings of these tubes. It is particularly recommended that they be consulted in the event it is proposed to design new equipment in which tubes will run at values greater than 85% of maximum under normal conditions.

Receiving-Inspection—All ITT tubes are x-rayed and are tested immediately prior to shipment. Despite the great care exercised in the design of the shipping containers and the care in selecting responsible freight forwarders, shipping damage does occasionally occur. Accordingly, all tubes should be inspected for both visible and invisible shipping damage immediately upon receipt. Any damage to the outer shipping container should be noted on the express receipt before it is signed. The container should be opened and both the tube and the inside of the container should be examined for evidence of any excessive rough handling during shipment.

Once it has been ascertained that no visible shipping damage is present, the tube should be installed in the equipment for which it is intended and operated after having followed the recommended break-in procedure. Meter readings and voltage settings should be compared with those experienced on tubes used previously. Any serious abnormalities should be discussed with your ITT field engineer to assess whether or not the tube may have hidden shipping damage. Filament current should be monitored with rated filament voltage applied. A clamp-on type ammeter is useful for this purpose if no filament current metering is included in the equipment instrumentation. The filament current should be within ±5 per cent of the rated value. In cases where there is a question of meter accuracy, it should be within ±5 per cent of that monitored on previous tubes. This is an important check because an abnormal reading can signify that the filament has been damaged. The filament structure may have many parallel electrical paths and the filament will appear to be intact if only a perfunctory check or an ohmeter continuity check is performed.

A shipping damage claim should be filed with the delivering carrier immediately upon discovering visible or hidden shipping damage. In order for the carrier to be able to honor such a claim, it must be filed within 7 days after receipt of the tube. Particular attention should be paid to hidden filament damage. Many equipments operate tubes very conservatively and tubes with this type of damage can often operate satisfactorily in such equipment for several hundred hours. Eventually, the portion of the filament which remained intact fails due to overheating. Many times the overheating causes the filament to bow and touch the grid.

Installation—New equipment should be designed with adequate clearance in the cabinet so that heavy tubes can be handled safely during installation and removal. On occasions where this has not proved practical, many astute manufacturers have provided their customers with special tools to lift particular tubes by the anode to facilitate safe installation.

Equipment manuals should stress periodic cooling system checks and an especially thorough check when a new tube is installed. Cleanliness of the envelope to insure high voltage holdoff and of connector surfaces to assure good electrical contact should also be emphasized. An undue amount of leverage should be avoided in tightening connectors.

Break-in Procedure—New tubes should be operated at normal rated filament voltage only, for thirty minutes before any other voltages are applied. Plate voltage should then be applied at the lowest possible value and increased gradually in steps over a period of an hour until the normal level is reached. This procedure will vary for individual tube types. Experience will determine if it may be shortened for smaller tubes or should be lengthened for larger tubes in a given application.

Storage—Tubes should be stored in a vertical position with the anode end down. They should be protected from extremes of heat and cold as well as from physical abuse. Every three months they should be given the break-in procedure outlined above and then operated at normal plate voltage for one hour.

Filament Care—The cathode of most of the tubes listed in this catalog is a structure of thoriated tungsten wire. There are a few older types listed which are still manufactured using bright tungsten cathodes. For the most part, the latter are made for replacement purposes only and, therefore, the comments here will be confined to thoriated tungsten cathodes.

Power Tube Applications

All of these cathodes are directly heated thus the cathode is really a filament, and the words filament and cathode are used interchangeably herein.

When a tube reaches a natural end-of-life condition it is usually because the emitting properties of the cathode have been exhausted. The life of this cathode and, therefore, the life of the tube can be extended many times beyond the normal 1000 hour warranty period if proper care is taken in the design of the equipment and if proper instructions on tube usage are passed along to the equipment users. Primary considerations in this respect are as follows.

1. **Filament starting**—The ratio of the hot to cold resistance of thoriated tungsten wire is approximately 10 to 1. Therefore, on a typical tube like the F-6696A, the nominal hot resistance would be 63.4 milliohms and the cold resistance would be 6.3 milliohms. If the rated voltage of 13 volts is applied to the cold filament without limiting current, the initial surge currents would be approximately 2000 amperes. Excessive starting current may easily warp or break a filament as the mechanical stress on the structure as a result of the magnetic field produced by this current is proportional to I_f^2.

   Consequently, current flow through a cold filament should be limited to 150% of the normal operating value for large tubes and 250% for medium types unless otherwise specified on a particular tube data sheet. The three most common methods of limiting this current are a high reactance transformer, manual control of the filament voltage either through a multi-tapped transformer or an auto transformer, or by means of a series of resistances in the filament circuit which are shorted out one at a time until rated voltage and current are achieved. ITT applications engineers can supply useful details on any of these methods.

2. **Maintaining proper voltage**—It is essential that the filament voltage on thoriated tungsten filaments be held to within $\pm 5\%$ of the rated value. Operation at high voltages will destroy the layer of tungsten carbide which protects the emitting layer and operation at low voltages will destroy the emitting layer itself either by shutting off the supply of thoria, or by reducing the space charge and thus subjecting the filament to increased ion bombardment. All transmitters or industrial heating equipment should include filament voltage metering. Not quite as essential but highly desirable, are filament current meters. The ability to monitor filament current is useful in performing a receiving inspection on new tubes, in performing filament processing when required and in predicting when a tube is approaching end-of-life condition.

3. **Filament recovery and processing**—Occasionally, a thoriated tungsten cathode which appears to have lost its emissive capabilities may be reactivated by applying filament voltage *only* in accordance with one of the following schedules.

   A. Apply 110% of rated value of filament voltage for a few hours or over night.

   B. If the emission fails to respond after schedule A, run at 30% above normal voltage for 10 minutes, then at 10% above normal for 20 to 30 minutes.

   C. In extreme cases, where A and B have failed to give results, and at the risk of burning out the filament, run at 75% above normal for 3 minutes followed by schedule B.

   This procedure is not effective in cases where the protective layer of tungsten carbide formed on the surface of the filament wire during manufacture has been severely depleted. This layer occupies a portion of the area cross-section of the wire and is relatively nonconductive. Its depletion is effected by reduction of the tungsten carbide back into tungsten which is a better conductor. Therefore, the conductance of the wire is effectively increased by decarburization. Accordingly, filament decarburization is accompanied by a rise in filament current. When filament current at rated filament voltage has risen to a value, 10% above that originally recorded at rated filament voltage, the tube is at or near end of life.

Handling—Tubes should be protected from shock and vibration during handling. Some tubes are quite heavy and require two people or proper materials handling equipment to transport them safely. They should always be handled by the anode and never by the envelope which would put an unreasonable amount of strain on the ceramic to metal or glass to metal seals.

Cleanliness—The equipment users should be encouraged to keep the tubes clean in order to encourage long life. A periodic maintenance check which includes cleaning the radiator fins on forced air cooled tubes and the anode itself on water or vapor cooled tubes will promote long life through improved heat transfer. Cleaning of non-glazed ceramic envelope areas with ordinary household cleanser or of glass or glazed ceramic envelopes with carbon tetrachloride to remove fingerprints and all other foreign matter will improve the insulating properties of these materials. This is extremely important as foreign matter on these surfaces will increase the probability of external arcing which could puncture the tube at the seal areas.

Fault Protection—High voltage arcing, whether internal or external, will destroy or impair a high vacuum tube. It is recommended that electronic crowbar circuit protection devices be installed in all high power equipment using these tubes and especially in those applications where such arcing is likely due to conditions of irregular line or load variations. Ball gaps or inert gas-filled spark gaps should be used across the tube terminals to protect against external arcing. ITT applications engineers can be of valuable assistance in providing information upon any aspect of tube protection.

ELECTRON TUBE DIVISION ITT
P.O. Box 100 Easton, Pennsylvania 18042

IMPORTANT
Make payments to Lepel High Frequency Laboratories, Inc., Maspeth, N. Y. For additional terms and conditions see back of acknowledgment copy.

LEPEL HIGH FREQUENCY LABORATORIES, INC.

59-21 QUEENS - MIDTOWN EXPRESSWAY • MASPETH, NEW YORK CITY, N. Y. 11378

(212) 426-4580

PACKING LIST

FACTORY ORDER NO. 37922

F.O. NUMBER	DATE REC'D.	CUSTOMER'S PURCHASE ORDER NUMBER	SHIP WITH F.O. NUMBER	INVOICE DATE	INVOICE NO.
37922	8/20/74	Y 9019			37922

SHIP TO (SAME AS SOLD TO UNLESS OTHERWISE INDICATED):
BRIGHAM YOUNG UNIVERSITY
RECEIVING DEPT
685 EAST 1700 NORTH
PROVO, UTAH 84601

SOLD TO:
BRIGHAM YOUNG UNIVERSITY
FINANCIAL SERVICES
D-148 ABRAHAM SMOOT BUILDING
PROVO, UTAH 84601

TAXABLE		
YES	X	NO
INSPECTION AT:	LEPEL	DESTINATION
VOLTAGE	CY	PH
460	60	3
DEL. REQ.	DEL. PROM.	
ASAP		

IF SHIPMENT IS INCORRECT, PLEASE NOTIFY US IMMEDIATELY. NO CLAIMS ALLOWED AFTER 10 DAYS.

SEE BELOW FOR CONDITIONS RELATING TO GOODS DAMAGED IN TRANSIT.

[redacted]

ROUTING	BACK ORDER FROM	BACK ORDER NO.	INSTALLATION CERTIFICATE

TERMS: NET 30 FROM INVOICE DATE	F.O.B. LEPEL PLANT	TRANSPORTATION COMPANY	COL.	PPD.	FGT. CHG.	WAYBILL NO.	NO. OF PKG.
							3

ORD.	SHIP.	B.O.	MODEL NO.	SERIAL NO.	DESCRIPTION	UNIT PRICE			TOT. AMOUNT DUE
1	1		T-10-3-KC-F1	7416 RP	HIGH FREQUENCY INDUCTION H'T'G UNIT: W/THYRATRON REGULATOR DWG NO: T-1001-237 (INCLUDING):	[redacted]	[redacted]		
1	1		018-0005	E4-0255	TUBE, OSCILLATOR (5606)				
3	3		018-0025		TUBE, THYRATRON (678)				10689.00
3	3		018-0020		TUBE, RECTIFIER (575A)				
1	1		007-0022	77464	CAPACITOR, TANK (.0076UF)			DISCOUNT	1189.00
2	2		038-0019		RESISTOR, GRID (15,000 OHMS, 200W)				
1	1		019-0002		STRAINER, WATER (1" PIPE)				
2	2		T111A		LOAD LEADS (COPPER-RIGID) 10" LONG				
1	1				LOAD COIL				
1	1				INSTALLATION DRAWING				9500.00
1	0	1			INSTRUCTION MANUAL				
1	1		049-0157	7477	LOAD COIL TRANSFORMER				
1	1		001-0003		ADAPTOR (12:1)				
1	1		001-0004		ADAPTOR (12:3)				
1	1				DWG. NO. 335-0013				

PACKED BY N396

CONSIGNEE TAKE NOTICE - UNPACK AND EXAMINE THIS SHIPMENT PROMPTLY

This merchandise was carefully inspected, tested and packed and left our possession in perfect condition. Any irregularity present on receipt, occurred in transit, and since the merchandise has been shipped f.o.b. shipping point, our responsibility for the shipment ceased upon delivery to the carrier.

External evidence of damage must be noted on freight bill or express receipt and signed by carrier's agent. Failure to do so may result in carrier refusing to honor your claim.

Concealed damage means that which does not become apparent until merchandise has been unpacked and tested. The contents may be damaged in transit even though the carton shows no external evidence. If any irregularity is found, submit a written request to carrier for an inspection, and hold merchandise and its container for the carrier's inspector.

If the above instructions are complied with, we shall cheerfully render all possible assistance in establishing claims against the transportation companies for the loss or damage in transit, but this cooperation on our part does not make us responsible for collection or replacement of the material.

Where necessary, replacements will be shipped and invoiced which then will constitute part of your claim.

Return no merchandise to the factory until we have authorized such return and forwarded shipping instructions.

When merchandise is shipped f.o.b. destination, notice of damage should be given to Lepel immediately upon receipt of goods.

LepeL HIGH FREQUENCY LABORATORIES, INC

59-21 QUEENS-MIDTOWN EXPRESSWAY • MASPETH, NEW YORK CITY, N.Y. 11378 • TEL. (212) 426-4580 • CABLE: LEPELA

PHOTO BULLETIN NO. 4

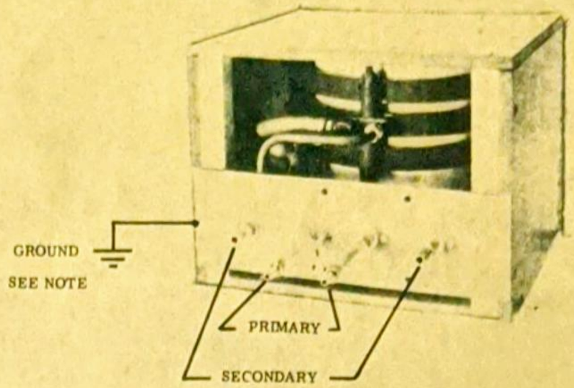
LOAD COIL TRANSFORMER (LCT-4)

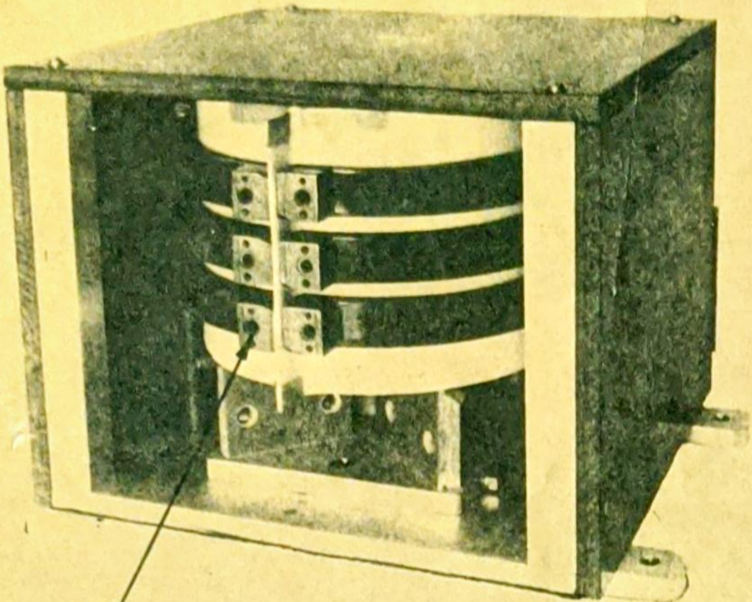
STOCK NO. 049–0157

(FOR USE WITH 2½ TO 50KW 3Ø KC GENERATOR)

APPROX. DIM: 10 1/2" HEIGHT, 13" WIDTH, 13" DEPTH
PLUS REAR LEADS

REAR VIEW





NATIONAL 'O' RINGS NO. 622705
8 REQUIRED

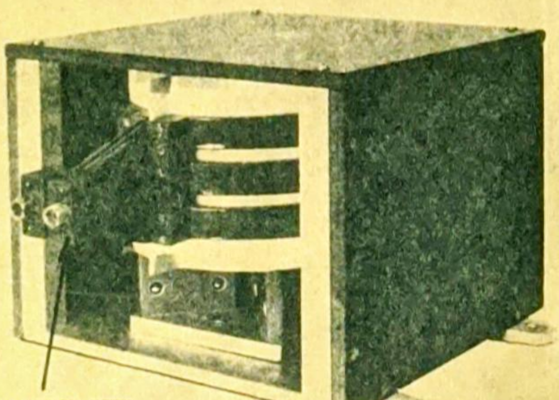
FRONT VIEW

NOTE: WHEN A LOAD COIL WITH COAXIAL CABLE IS USED
- A - REMOVE EXISTING GROUND STRIP
- B - ISOLATE THE TRANSFORMER BASE PLATE FROM GROUND

PRIMARY CONNECTIONS:
1 - CONNECT TO GENERATOR R.F. OUTPUT TERMINALS FOR PRIMARY POWER AND WATER COOLING
2 - CONNECTORS SUPPLIED:
- A - MALE CONNECTOR 1/4" NPT X 3/8" FLARE
- B - TUBE ADAPTOR 3/8" FLARE X 1/4" COMPRESSION

SECONDARY CONNECTIONS:
1 - CONNECT TO SEPARATE WATER SUPPLY FOR SECONDARY AND LOAD COIL COOLING
2 - CONNECTORS SUPPLIED:
- A - HOSE NOZZLE 1/4" NPT X 3/8" HOSE

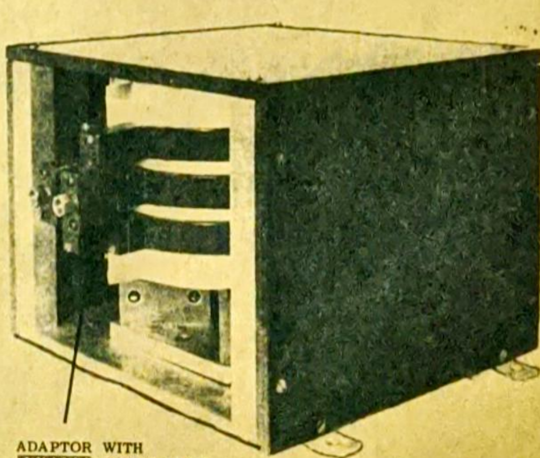
LCT-4 WITH 12 TO 1 ADAPTOR
ADAPTOR STOCK NO. 001-0003



ADAPTOR WITH INVERTED FLARE MALE CONNECTOR 1/4" TUBE

KEEP LOAD COIL LEADS AS SHORT AS POSSIBLE

LCT-4 WITH 12 TO 3 ADAPTOR
ADAPTOR STOCK NO. 001-0004



ADAPTOR WITH INVERTED FLARE MALE CONNECTOR 1/4" TUBE

Regional Office
Tel. (714) 540-1650

17815 Sky Park Circle
P.O. Box 4535
Irvine, Calif. 92664

LepeL High Frequency Laboratories, Inc.

Manufacturers of Induction Heating Equipment

Main Office
59-21 Queens-Midtown Expressway
Maspeth, N.Y.C. 11378
Tel. (212) 426-4580

ROBERT B. KENNEDY
Service Engineer

REF DESIG	LEPEL STOCK NO.	DESCRIPTION	SPECIFICATION	ZONE
RC1	[illegible]	THYRATRON REGULATOR T-[illegible]		[illegible]
RC2	[illegible]	THYRATRON REGULATOR T-[illegible]		
B1	[illegible]	[illegible], MOTOR	[illegible]	
B2	[illegible]	[illegible], MOTOR	[illegible]	
C1	[illegible]	CAPACITOR	[illegible]	
C2	[illegible]		[illegible]	
C3	[illegible]		[illegible]	
C12	[illegible]		[illegible]	
C13	[illegible]			
C14	[illegible]		[illegible]	
C15	[illegible]		[illegible]	
C7	[illegible]		[illegible]	
C8	[illegible]		[illegible]	
C9	[illegible]		[illegible]	
C10	[illegible]		[illegible]	
C11	[illegible]		[illegible]	
C16	[illegible]		[illegible]	
C17	[illegible]	CAPACITOR	[illegible]	
CB1	[illegible]	CIRCUIT BREAKER	[illegible]	
DS1	[illegible]	LAMP, PILOT	[illegible]	
DS2	[illegible]		[illegible]	
DS3	[illegible]		115 V, 6 W	
DS4	[illegible]		[illegible]	
DS5	[illegible]		6.3 V	
DS6	[illegible]		115 V, 6 W	
DS7	[illegible]		115 V, 6 W	
DS8	[illegible]	LAMP, PILOT	115 V, 6 W	
E1		SPARK GAP	SUPPLIED WITH C1	
E2		SPARK GAP	SUPPLIED WITH C2	
E3		SPARK GAP	SUPPLIED WITH C3	
E4	[illegible]	PROTECTOR, [illegible]	DWG # 52800-16	
F1	[illegible]	FUSE	[illegible]	F-3
F2	[illegible]		[illegible]	
F3	[illegible]		[illegible]	
F4	[illegible]	FUSE	[illegible]	

REF DESIG	LEPEL STOCK NO.	DESCRIPTION	SPECIFICATION	ZONE
J1	[illegible]	RECEPTACLE	[illegible]	
K1	[illegible]	[illegible]	[illegible]	
K2	[illegible]	[illegible]	[illegible]	
K3	[illegible]	[illegible]	[illegible]	
K4	[illegible]	[illegible]	[illegible]	
K5	[illegible]	[illegible]	[illegible]	
K6	[illegible]	[illegible]	[illegible]	
K7	[illegible]	[illegible]	[illegible]	
K8	[illegible]	[illegible]	[illegible]	
K9	[illegible]	[illegible]	[illegible]	
K10	[illegible]	[illegible]	[illegible]	
L1	[illegible]	CHOKE, RF	[illegible]	
L2	[illegible]	[illegible]	[illegible]	
L3	[illegible]	COIL, [illegible]	[illegible]	
L4	[illegible]	COIL, TANK	[illegible]	
L5	[illegible]	SOLENOID, VALVE	[illegible]	
M1	[illegible]	KILOVOLTMETER	[illegible]	
M2	[illegible]	VOLTMETER [illegible]	[illegible]	
M3	[illegible]	[illegible]	[illegible]	
M4	[illegible]	[illegible]	[illegible]	
M5	[illegible]	METER, [illegible]	[illegible]	
R1	[illegible]	RESISTOR, [illegible]	3 MEG Ω	
R2	[illegible]	, [illegible]	6 MEG Ω	
R3	[illegible]	, [illegible]	6 MEG Ω	
R4	[illegible]	, GRID	15 K Ω, 200 W	
R5	[illegible]	, GRID	15 K Ω, 200 W	
R6	[illegible]	, [illegible]	250 Ω, 25 W	
R7	[illegible]	RESISTOR, POT	1 K Ω, 2 W	
S1	044-0062	SWITCH, THERMO	[illegible]	D-1
S2	044-0056	, WATER FLOW	SPDT-4 GPM	D-3
S3	044-0073	, PB-STOP	INC. RED	D-7
S4	044-0072	, PB-START	IND. GREEN	D-7
S5	044-0064	[illegible]	N.O. [illegible]	D-3
S6	044-0012	, TOGGLE	4PDT, 15 AMP, 125 VAC	D-2
S7	044-0013	, TOGGLE	DPDT, 10 AMP, 250 VAC	D-2
S8	044-0078	SWITCH, TOGGLE	4PDT, 15 AMP, 125 VAC	C-4, 3-5

REF DESIG	LEPEL STOCK NO.	DESCRIPTION	SPECIFICATION	ZONE
S9	[illegible]	SWITCH, PB-STOP	INC. RED	C-4
S10	[illegible]	, PB-START	IND. GREEN	C-4
S11 THRU S15 INCL.	[illegible]	, INTLK	[illegible]	[illegible]
S16	[illegible]	[illegible]	[illegible]	[illegible]
S17	[illegible]	, TOGGLE	[illegible]	[illegible]
S18	[illegible]	, SELECTOR	[illegible]	[illegible]
S19	[illegible]	, PB-STOP	INC. RED	[illegible]
S20	[illegible]	, PB-START	IND. GREEN	[illegible]
S21	[illegible]	SWITCH, [illegible]	[illegible]	[illegible]
T1	[illegible]	XFMR, [illegible]	[illegible]	[illegible]
T2	[illegible]	[illegible]	[illegible]	[illegible]
T3	[illegible]	[illegible]	[illegible]	[illegible]
T4	[illegible]	XFMR, [illegible]	[illegible]	[illegible]
TB1	[illegible]	TERMINAL BOARD	[illegible]	
TB2	[illegible]	[illegible] STRIP		
V1	[illegible]	TUBE, THYRATRON	[illegible]	[illegible]
V2	[illegible]	, THYRATRON	[illegible]	[illegible]
V3	[illegible]	, THYRATRON	[illegible]	[illegible]
V4	[illegible]	, OSCILLATOR	[illegible]	[illegible]
V5	[illegible]	, RECTIFIER	[illegible]	[illegible]
V6	[illegible]	, RECTIFIER	[illegible]	[illegible]
V7	[illegible]	TUBE, RECTIFIER	[illegible]	[illegible]
W1	043-0053	CABLE ASSY	[illegible]	[illegible]
XDS1	[illegible]	SOCKET, PILOT LAMP	RED LENS	
XDS2	[illegible]		RED LENS	
XDS3	[illegible]		AMBER LENS	
XDS4	[illegible]		RED LENS	
XDS5	[illegible]		RED LENS	
XDS6	[illegible]		GREEN LENS	
XDS7	[illegible]		AMBER LENS	
XDS8	[illegible]	SOCKET, PILOT LAMP	RED LENS	
XF1	[illegible]	FUSEHOLDER	[illegible]	
XF2	[illegible]		[illegible]	
XF3	[illegible]		[illegible]	
XF4	[illegible]	FUSEHOLDER	[illegible]	

ITEM	QTY	PART NO.	DESCRIPTION	SPECIFICATION

LIST OF MATERIALS

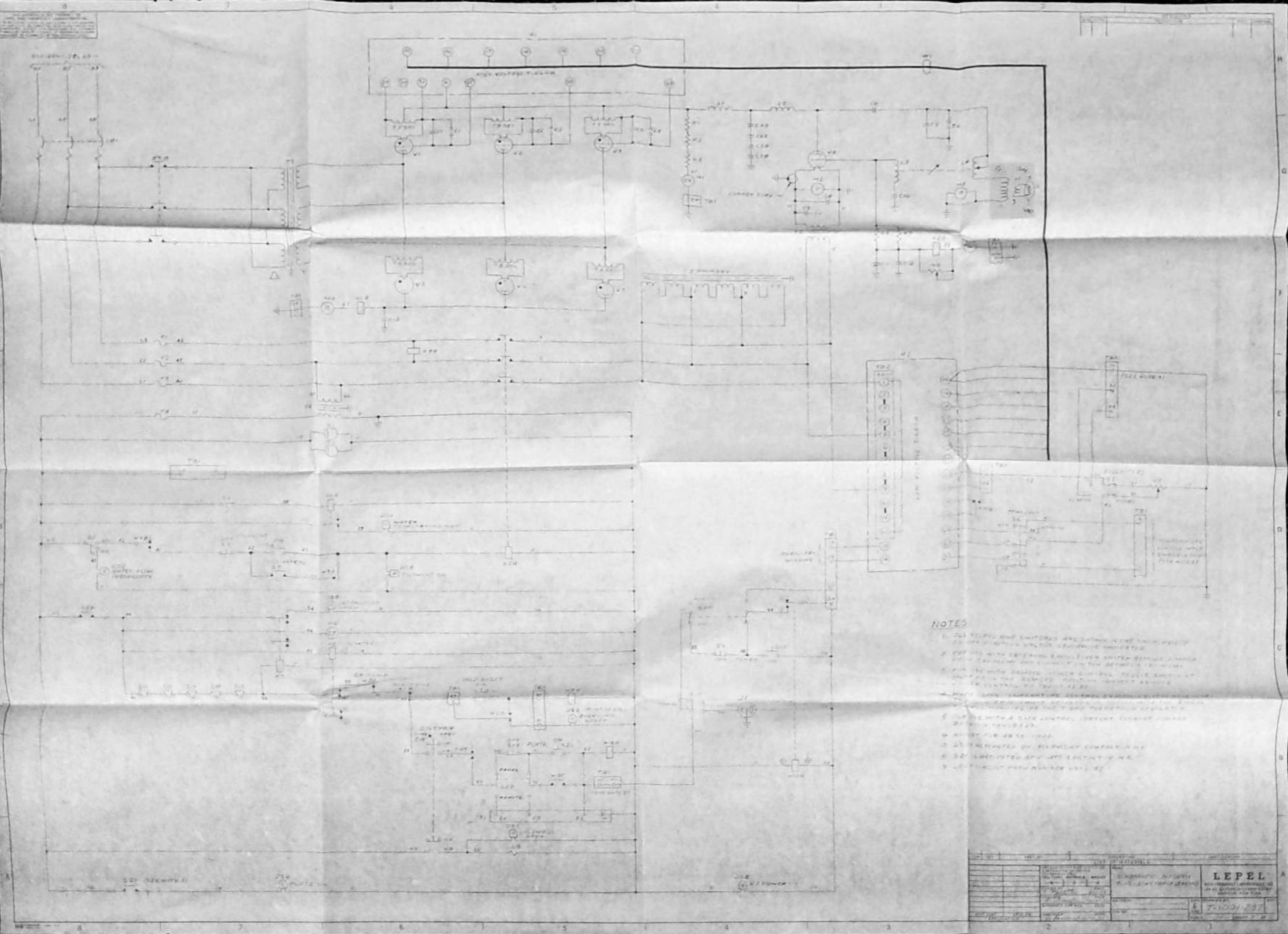
NAME: SCHEMATIC DIAGRAM [illegible]

LEPEL

DRAWN BY: [illegible] DATE: [illegible]

SIZE: D DRAWING NO. T-1001-237

SHEET 2 OF [illegible]


NOTES
LEPEL